

Arnav Vikram Shinde

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📍 Indore, M.P.

EDUCATION

Mechanical Eng. Shri G.S. Institute of Technology And Science	06/2027 Indore, M.P.
12th Vidya Bhavan	03/2023 Ujjain, M.P.
10th Christu Jyotu Convent School	03/2021 Ujjain, M.P.

SKILLS

CAD

Creo, SolidWorks, Solidedge, NX CAD

CAE/FEA

Ansys Mechanical/Workbench, Modal, Random Vibration, Thermal, Static Structural, Contact Mechanics, Rigid Body Dynamics

Multi Body Dynamics

MSC Adams, IPG CarMaker, Ride Analysis, Fourpost Rig Simulation, Drop Test

EXPERIENCE

Intern Gabriel India Ltd. 	05/2025 – 07/2025 Dewas, M.P.
- Acquired knowledge of suspension systems and damper functionality. - Studied supply chain management principles and manufacturing line operations & gained expertise in Design Verification Plan (DVP) and Design Failure Mode and Effects Analysis (DFMEA).	
Intern Avinya MilTech 	06/2026 – 07/2026 Indore
- Developed precision engineering drawings and 3D models of optical and mechanical components in Solid Edge, including a military-grade, waterproof enclosure for a Jetson Nano Orin AI camera system integrating LiDAR, IR, and HD imaging modules. - Performed CAE modal, random vibration, and thermal analysis on a satellite-related assembly. - Contributed to developing a complete kinematic simulator (Creo, MSC ADAMS) for a CMP lens-polishing machine—designing a quick-return-free four-bar oscillating arm, deriving disc profile geometry, calculating material removal rates using Preston's equation, developing a predictive re-flattening algorithm, and selecting bearings to optimize machine life—resulting in an app that accurately predicts machine behavior.	
Team Lead Team GSRacers BAJA 	03/2025 – 02/2026 Indore, M.P.
- Served as team's Technical Head, Vehicle Dynamics Head, Virtual Dynamics for 2026 e-Baja event. - Guided AIR1 to the Virtual Dynamic department.	

PROJECTS

Reaction Wheel Attitude Control System

- Designed and CAD-modeled a reaction-wheel ACS for a nanosatellite, deriving torque/momentum budgets and moment-of-inertia calculations from rigid-body attitude dynamics
- Built an MBD simulation of the attitude control loop and validated simulated maneuver response against theoretical predictions
- Performed Modal and Random Vibration Analysis in ANSYS to verify structural integrity and natural frequencies under launch loads

Multi-Stage Gear Train Dynamics

- Developed and evaluated a multi-stage gear train model in MSC Adams, incorporating gear-to-gear contact interactions and rotational motion to study torque transmission, contact forces, and transient dynamic response under operating conditions.
- Analysed time-history data of transmitted contact torque to evaluate load fluctuations, gear engagement behaviour, and overall drivetrain dynamic performance.

Sinusoidal Ride Analysis

- Performed ride dynamics simulations of the Baja vehicle over uneven and sinusoidal off-road terrain inputs to evaluate vertical response, suspension articulation, body motion, and wheel movement under dynamic conditions.
- Assessed ride stability and suspension effectiveness, focusing on terrain disturbance absorption and maintaining consistent wheel-ground contact across varying inputs.

IPG CarMaker — Virtual Dynamics Simulation

- Developed and validated a complete vehicle model in IPG CarMaker, simulating diverse tracks and off-road terrains to evaluate suspension behaviour, steering response, handling, stability, ride dynamics, and overall vehicle performance. Used simulation results to identify limitations and iteratively optimize vehicle setup and design for improved dynamic performance.
- Led the Virtual Dynamics department (IPG CarMaker) at Team GSRacers, driving simulation-based vehicle development and performance optimization, contributing to the team achieving the #1 rank across India.

Four-Post Rig Ride Simulation

- Conducted a four-post road-load simulation of the Baja vehicle under controlled vertical inputs to evaluate suspension kinematics, wheel displacement, tire-ground interaction, load transfer, and overall chassis response.
- Analysed suspension travel and dynamic load response to assess the vehicle's ability to absorb and isolate road-induced loads while maintaining stable wheel-ground interaction.

AWARDS

AIR 1 in Virtual Dynamics (Validation Event) E-BAJA 2026

01/2026

SAE BAJA

AIR 7 overall in M-BAJA 2024, AIR 3 in Physical Dynamics in M-BAJA 202

01/2024

SAE BAJA